INCH-POUND
MIL-PRF-32497
15 May 2014

PERFORMANCE SPECIFICATION

REFRIGERATOR, BIOMEDICAL, FOR SURFACE SHIPS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers U.S. Navy shipboard biomedical refrigerators for storage of medicines and vaccines on surface ships.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-S-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)
MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-1399-300	-	Electric Power, Alternating Current

(Copies of these documents are available online at http://quicksearch.dla.mil.)

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

BUREAU OF MEDICINE AND SURGERY (BUMED)

BUMEDINST 6230.15 - Immunizations and Chemoprophylaxis for the Prevention of Infectious Diseases

(Copies of this document are available online at www.med.navy.mil.)

CENTERS FOR DISEASE CONTROL AND PREVENTION (CDC)

Vaccine Storage and Handling Toolkit

(Copies of this document are available online at www.cdc.gov/.)

MILITARY VACCINE (MILVAX) AGENCY

Vaccine Storage and Handling Guidelines

(Copies of this document are available online at https://www.vaccines.mil/.)

NAVAL SURFACE FORCES COMMAND INSTRUCTIONS

COMNAVSURFPACINST 6000.1/ COMNAVSURFLANTINST 6000.1 - Shipboard Medical Department Procedures Manual

(Copies of this document are available online at https://cpf.portal.navy.mil/sites/cnsp/Pages/Directives.aspx.)

OPNAV INSTRUCTIONS

OPNAVINST 5090.1 - Environmental Readiness Program

(Copies of this document are available online at http://doni.documentservices.dla.mil/default.aspx.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM F2520 - Standard Specification for Reach-in Refrigerators, Freezers, Combination Refrigerator/Freezers, and Thaw Cabinets

(Copies of this document are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Units furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.4).

3.2 Design. Unless otherwise specified (see 6.2), units shall be constructed in accordance with ASTM F2520 design and material details, including the supplemental requirements for naval shipboard use. Units shall be designed with minimal construction, installation, and maintenance requirements. Units shall be designed for under-counter or upright use as specified (see 6.2). Design characteristics shall be verified in accordance with 4.4.

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3.2.1 Access. Standard units shall pass through a 26-inch (660-millimeter) wide by 66-inch (1676-millimeter) shipboard hatch without major disassembly in accordance with ASTM F2520, including the supplemental requirements for naval shipboard use. When required, units may be of the modular type, capable of being disassembled and reassembled at location of installation (see 6.2).

3.2.2 Refrigerant. Unless otherwise specified (see 6.2), units shall use R134A or R404A. Refrigerants shall be non-ozone-depleting and readily available, obtainable, and allowable onboard in-service ships, in accordance with OPNAVINST 5090.1.

3.2.3 Condenser. Units shall have an air-cooled condenser constructed of either copper, aluminum, or steel in accordance with ASTM F2520, including the supplemental requirements for naval shipboard use.

3.2.4 Condensation. Units shall utilize forced air evaporators to remove condensate in accordance with MILVAX Vaccine Storage and Handling Guidelines. Units shall provide thermostatically controlled vaporizer pans as means of disposing evaporator runoff in accordance with ASTM F2520, including the supplemental requirements for naval shipboard use.

3.2.5 Circulation. Units shall contain a front-facing circulation fan to assist in proper operation within the provided installation clearances.

3.2.6 Internal shelving. Units shall contain adjustable and removable shelving. Shelving material shall be the manufacturer's standard product shelving, with the exception that glass shall not be used as a shelving material.

3.2.7 Internal lighting. Units shall contain internal lighting for ease of visibility of product. Lighting shall be located within the cabinet and shall be recessed and protected by a cover.

3.2.8 Security. Units shall be provided with a single-hinged hasp-style fastening latch installed on the door for use with a padlock to secure product. Latch shall be constructed of stainless steel in accordance with ASTM F2520, including the supplemental requirements for naval shipboard use.

3.3 Storage characteristics.

3.3.1 Capacity. Units shall have the capacity to maintain the inventory required for mission support and to facilitate good air circulation around the product. Units shall have a minimum capacity of 5 cubic feet (142 liters), with additional models available in increasing increments of 5 cubic feet (142 liters) as needed (see 4.4 and 6.2).

3.3.2 Internal clearances. Units shall allow for an additional 3 inches (77 millimeters) of open space around the perimeter of the refrigerated product to allow for proper air circulation in accordance with OPNAVINST 5090.1. Clearances shall be verified in accordance with 4.4.

3.4 Performance characteristics.

3.4.1 Environmental conditions. Units shall be able to operate satisfactorily and maintain internal temperatures as identified in 3.4.6 with an outside ambient temperature of up to +120 °F (+49 °C) (see 4.5.1).

3.4.2 Shock. Units are typically installed in manned spaces aboard ships and therefore shall be Grade B shock qualified in accordance with MIL-S-901 (see 4.5.2).

3.4.3 Inclined operation. Units shall operate satisfactorily when inclined at an angle of 15 degrees on each side of the vertical in each of two vertical planes at right angles to each other, with no abnormal variations in temperature or performance for which the unit is designed in accordance with ASTM F2520, including the supplemental requirements for naval shipboard use (see 4.5.3).

3.4.4 Water resistance. Electronic components shall be water-resistant to maintain proper operation in the event of splashing or exposure to water. Water resistance shall be verified in accordance with 4.5.5.

3.4.5 Electromagnetic interference (EMI) considerations. Units shall meet all EMI requirements for mission essential, non-spectrum dependent, protected below deck equipment in accordance with MIL-STD-461 (see 4.5.4).

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3.4.6 Internal temperature range. Refrigerator units shall be able to maintain an internal temperature range of 35 to 46 °F (2 to 8 °C). Set temperatures shall be in accordance with BUMEDINST 6230.15, COMNAVSURFPACINST 6000.1/COMNAVSURFLANTINST 6000.1, and the CDC Vaccine Storage and Handling Toolkit. In the event of power loss, units shall maintain acceptable internal temperature ranges for a minimum of 6 hours at an ambient temperature of 77 °F (25 °C). Temperature range shall be verified in accordance with 4.5.5.

3.4.7 Temperature measurement. Units shall be equipped with a calibrated digital thermometer. Temperature measurement capability shall be verified in accordance with 4.5.5.

a. Thermometers shall be able to be recalibrated according to manufacturer specifications.

b. Digital thermometers shall be capable of accurately reporting temperatures within ±1 °F (±0.5 °C).

c. Temperature measuring and monitoring equipment shall be capable of sustaining full operational capability for a minimum of 6 hours in the event of primary power loss.

3.4.8 Temperature logging. Units shall have the ability to log the time and duration of power loss, loss of temperature range, and temperature periodically using a digital temperature logger. The temperature logger shall include an adjustable sampling/logging interval. The smallest sampling interval shall be no greater than 60 seconds. Temperature logger shall have the storage capacity to store a minimum of 7 days of samples at the shortest sampling interval. Logs shall be transferrable and viewable on external systems via a USB interface for more detailed viewing or sharing. Temperature logger shall be capable of sustaining full operational capability for a minimum of 6 hours in the event of primary power loss. Temperature logging capabilities shall be verified in accordance with 4.5.5.

3.4.9 Control interface. Units shall use a digital system for controlling temperature and alarm set-points. The digital system shall display the current temperature setting and options to view temperature logs. Digital temperature controllers shall be capable of maintaining temperatures within a range of ±2 °F (±1 °C). Interface shall be evaluated in accordance with 4.4 and 4.5.5.

3.4.10 Alarms. Units shall have a combination audible and visual alarm system for both temperature ranges (high or low temperature exceeded) and loss of power. Alarms shall sound at the unit's location, as well as remotely in a continually manned space, upon power loss or when required temperature range is not maintained. Alarm set-points shall be fully adjustable by installation personnel in accordance with acceptable temperature ranges established in 3.3.1. Alarm set-points shall be able to be locked and adjustability shall be restricted to authorized personnel for post-installation adjustments. Alarm system shall be evaluated in accordance with 4.4.

3.5 Power characteristics.

3.5.1 Power compatibility. Units shall be compatible with 115-volt, 60-Hertz, single-phase, alternating current (AC) in accordance with ASTM F2520, including the supplemental requirements for naval shipboard use, and MIL-STD-1399-300. Power compatibility shall be verified in accordance with 4.4.

3.5.2 Surge protection. Units shall be protected from and remain operational after unexpected power surges and current excursions. Surge protection shall be verified in accordance with 4.4.

3.6 Installation.

3.6.1 Foundation. Units shall be provided with foundations or legs suitable for bolting to the deck during installation. Legs shall be a minimum 6 inches (153 millimeters) in height. Alternatively, foundations shall totally enclose the base of the unit and be a minimum 4 inches (102 millimeters) in height (see 4.4 and 6.2).

3.6.2 Clearances. Units shall be able to operate with minimal clearances required between unit and adjacent equipment or structures. Allowable range of clearance shall be 0 to 3 inches (0 to 77 millimeters). Operational clearances shall be verified in accordance with 4.5.5.

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3.7 Maintenance. Units shall be maintainable, repairable, and upgradeable by shipboard personnel, given installation requirements established in 3.6. Individual components shall be replaceable at shipboard O-level. Maintenance and repair access shall be from the front of the unit. Components shall be able to be quickly disconnected or units shall have the capability to slide the entire component module out of the unit for ease of repair or replacement. Maintainability shall be evaluated in accordance with 4.4.

3.8 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. Materials shall be evaluated in accordance with 4.4.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall include the examination of 4.4 and the tests of 4.5.1 through 4.5.5 (see table I). Qualification tests shall be conducted at a laboratory acceptable to the Naval Sea Systems Command (NAVSEA).

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4 and operational compliance inspection specified in 4.5.5 (see table I).

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TABLE I. Qualification and conformance inspections.

Characteristic	Qualification	Conformance	Requirement	Verification
Design	X	X	3.2	4.4
Capacity	X	X	3.3.1	4.4
Internal clearances	X	X	3.3.2	4.4
Control interface	X	X	3.4.9	4.4
Alarms	X	X	3.4.10	4.4
Power characteristics	X	X	3.5	4.4
Installation	X	X	3.6	4.4
Maintenance	X	X	3.7	4.4
Recycled, recovered, environmentally preferable, or biobased materials	X	X	3.8	4.4
Environmental conditions	X		3.4.1	4.5.1
Shock	X		3.4.2	4.5.2
Inclined operation	X		3.4.3	4.5.3
EMI considerations	X		3.4.5	4.5.4
Water resistance	X	X	3.4.4	4.5.5
Internal temperature range	X	X	3.4.6	4.5.5
Temperature measurement	X	X	3.4.7	4.5.5
Temperature logging	X	X	3.4.8	4.5.5
Control interface	X	X	3.4.9	4.5.5
Clearances	X	X	3.6.2	4.5.5

4.4 Examination. Units shall be subjected to a thorough inspection to confirm adherence to design characteristics as identified in 3.2, construction and installation requirements as identified in 3.2 and 3.6, dimensional requirements as identified in 3.3.1 and 3.3.2, interface requirements as identified in 3.4.9 and 3.4.10, power characteristics as identified in 3.5, maintenance requirements as identified in 3.7, and material requirements established in 3.8. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Methods of verification.

4.5.1 Shipboard environment. Units shall be tested under normal operating conditions in accordance with MIL-STD-167-1, Type I equipment, and 3.4.1. Units shall be secured to the test machine in the same manner that they will be secured on shipboard. Failure of the unit to perform its function during or after testing shall constitute failure of this test.

4.5.2 Shock. Adherence to shock requirements established in 3.4.2 shall be verified using test methods established in MIL-S-901.

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4.5.3 Inclined operation. Units shall be bolted to a test platform similar to shipboard installation and inclined at an angle of at least 15 degrees. The test platform shall replicate mounting requirements established in 3.6. Units shall be operated at 1 hour at each side of the vertical, in each of two vertical planes at right angles to each other. Any nonconformance with requirements established in 3.4.3 shall constitute failure of this test.

4.5.4 EMI. Adherence to EMI requirements established in 3.4.5 shall be verified using the following test methods established in MIL-STD-461: CE101, CE102, CS101, CS114, CS116, RE101, RE102, RS101, and RS103.

4.5.5 Operational compliance. Units shall conform to test methods established in ASTM F2520, including the supplemental requirements for naval shipboard use. Operational testing shall be performed to verify water resistance as identified in 3.4.4, temperature range and measurement as identified in 3.4.6 and 3.4.7, temperature logging and interface as identified in 3.4.8 and 3.4.9, and operational clearances as identified in 3.6.2.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This product is intended for use on U.S. Navy surface ships to provide reliable refrigeration to support required temperature range for stowage of medical cold chain products. Medical cold chain products such as vaccines are high-importance, high-cost items that require strict temperature regulation and alarm systems. Refrigerators in manned spaces on U.S. Navy surface ships are required to be qualified as shock Grade B to prevent hazards to personnel, a feature not available in commercial products. This product is not intended for use onboard submarines, or for products that require freezer storage.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Alternative design requirements (see 3.2).
c. Under-counter or upright unit (see 3.2).
d. Modular construction requirements (see 3.2.1).
e. Refrigerant requirements (see 3.2.2).
f. Cubic size of unit for required product to be stored (see 3.3.1).
g. Legs or foundation requirements (see 3.6.1).
h. Packaging requirements (see 5.1).
i. Data requirements (see 6.3).

6.3 Data requirements. When specified (see 6.2), illustrations should be provided detailing the installation methods of refrigerators. Illustrations should be provided showing the disassembly and reassembly of refrigerators if units are of the modular type.

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6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 32497 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.5 Subject term (key word) listing.

Cold chain
Medical
Medicine
R134A
R404A
Storage, medical
Vaccine

Custodians:
Army – GL
Navy – SH
Air Force – 99

Preparing activity:
Navy – SH
(Project 4110-2013-001)

Review activities:
Army – AV, CR4, MI, QM1
Navy – YD
Air Force – 03, 35
DLA – IS
GSA – FAS

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